EE337 Microprocessors Laboratory
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Problem set: 8 Date: March 16, 2026

Refer to Prof. Dinesh Sharma's slides and notes on serial I/O to do this lab. The link is given here: `https://ee337.github.io/dks/serial_io.html`.

1. [10 points] In this lab you will understand and use a program for communicating between Pt-51 and a computer using UART. This program will take inputs from the computer's keyboard that can be used in programs running on the kit, to perform appropriate operations.

   i) To connect the kit to a computer, you will be using the USB-to-UART adapter `Prolific PL2303` (Fig. 1) that was shipped along with the kit. The driver software for this adapter and the instructions for installing it can be found at the following link: `http://www.miklor.com/COM/UV_Drivers.php`
   After installing the software, connect the PL2303 adapter to one of the USB ports of your PC.

Figure 1: USB to UART adapter.

   ii) Connect the Pt-51 kit to the USB-to-UART adapter using F-F wires as described next. In the Pt-51 kit, port pin `P3.0` is the serial data input and `P3.1` is the serial data output. Connect `P3.0` of the kit to transmit data line (TxD) of the adapter. Connect `P3.1` of the kit to receive data line (RxD) of the adapter. Connect the `GND` pin of the kit to `GND` pin of adapter.

   iii) For recognizing keyboard inputs on the computer and transmitting to the kit, through the serial terminal, you need to use the `Realterm` software (or any equivalent software). This software will also be used to display the messages received from the Pt-51 kit on the PC. A screenshot of `Realterm` window is shown in Fig. 2.
   For Windows, download `Realterm` at: `https://realterm.i2cchip.com`
   For Linux, you can download `putty`.

   iv) Configure `Realterm` (or equivalent tool) to use the appropriate COM port and baud rate. This can be done by clicking on `Serial Port`/ `Port` tab and choosing appropriate

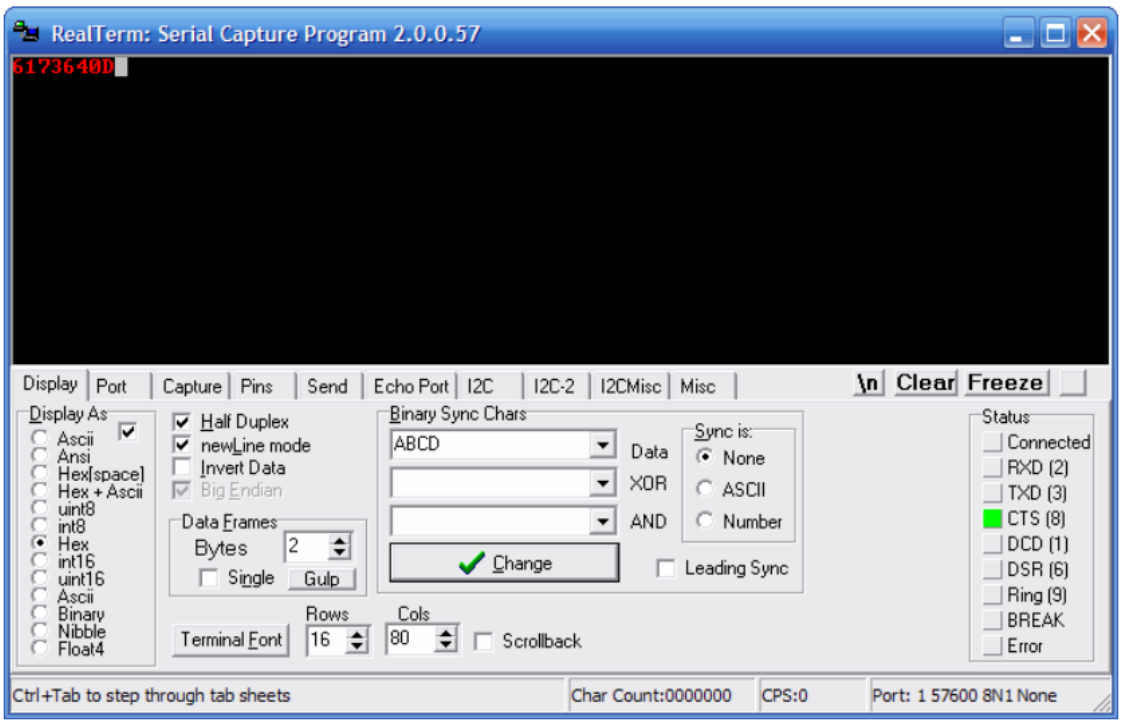

Figure 2: RealTerm: Serial/TCP Terminal

COM port as the port to which the USB-to-UART adapter is connected. Set the baud rate to 1200 (or that used in your program). Then click on `Open`.

With the kit connected to the computer using the USB-to-UART adapter, you will next write code to recognize key presses, perform an operation on the kit, and display messages sent by the kit on the computer.

Use `main.c` as starting (template) code. This uses `serial.c` and `lcd.h` for initialisation of UART and LCD, respectively. These files are part of the zip file shared with you.

The flow of the program is given below.

1. Complete the function `uart_init()` in `serial.c` to configure the serial port for UART communication. Use timer `T1` and a baud rate of 1200 bps.
2. The given `main.c` has code for transmitting a set of strings and to capture a key press on the PC. Compile the project, dump the hex file on to Pt-51 and run the program. Launch the `Realterm` window on your PC and you can see that the following is displayed on the window.

```
************************
******8051 Tests********
************************
Press 1 to test the LCD
Press 2 to test the LED
```

3. Now, Pt-51 should be respond to the key presses on the PC's keyboard. With `Realterm` window in focus (active window), any key press on PC's keyboard will be captured by it and transferred to Pt-51.

- When 1 is pressed, it will be detected by Pt-51 and you will see a message "LCD Ok" on the LCD screen and "LCD tested" display on the `Realterm` window. This happens because of the function `lcd_test` in `main.c`.
- When 2 is pressed, it should be detected by pt-51 and should glow all four LEDs and display "LED tested" on the `Realterm` window. For this to happen, you will have to update the function `led_test` in `main.c`
- When any other key is pressed, you will see the message "Incorrect key pressed" displayed on the `Realterm` window.

2. [10 points] In this question, you will write a program to implement a **Room Booking System** using UART communication between the Pt-51 kit and a computer. User interactions will be simulated using key presses on the PC keyboard through UART.

   There are **five rooms** available in the system. Each room is assigned a number as shown below:

Room Number	Room Name
1	LA001
2	LA002
3	LH101
4	LH102
5	Conference Room

   The system must operate as described below.

   1. **Room Selection State**
      - Upon reset, the kit should display the following message on the UART terminal:
        > "Select Available Room (1–5):"
      - If the user enters a number between **1 and 5**, the corresponding room is selected and the system moves to the Date Selection state.
      - If any other input is entered, the following message should be displayed:
        > "**Input Room ID**"
      - Only rooms that have at least one remaining unbooked date must be displayed.
      - If all dates (1–7) for a particular room are booked, that room must be removed from the list of available rooms.
      - The system should then return to the **Room Selection state**.
   2. **Date Selection State**
      - Once a valid room is selected, the following message should be displayed:
        > "Select Date (1–7):"
      - The available booking dates are from **1st February to 7th February**, mapped as follows:

Input	Date
1	1st February
2	2nd February
3	3rd February
4	4th February
5	5th February
6	6th February
7	7th February

      - If a valid date (1–7) is entered, that date should be considered **booked** and the system should proceed to the Login Credentials state.
      - If any other input is entered, display:
        > "**Input Date**"
      - Only the dates that are currently unbooked for the selected room must be displayed as available options.

- Once a date is successfully booked for a room, that date must be removed from the list of available dates for that room in future selections.
- The system should return to the **Date Selection state**.

3. **Login Credentials State**
   - The system should prompt the user to enter a Login ID followed by a Password through UART.
   - The following **three login credentials** are predefined in the program:
     - ID: `EE337 MICRO` Password: *"Last 4 digits of your roll number"*
     - ID: `EE230 ANALOG` Password: `1234`
     - ID: `EE344 EDL` Password: `6789`
   - If the entered ID and password do **not** match any of the above credentials, the following message should be displayed:
     > "**Invalid Login Credentials**"

   - The system should then return to the **Login Selection state**.
   - If the login credentials are valid, the system should proceed to the **Summary state**.
4. **Summary State**
   - Upon successful room selection, date selection, and valid login, the UART display should show the following message format:

     ```
     **BOOKING COMPLETE**
     "Room Name" is booked for "Date of Booking"
     for "Login ID (Lab Name)"
     ```
   - After displaying the summary message, the system must return to the **Room Selection State**.
   - The system must retain booking information so that already booked dates and fully booked rooms are not shown again.

Implementation Note:

- Booking information (room availability and date availability) must be maintained internally by the program.
- Students are expected to use appropriate data structures (arrays, flags, or state variables) to track bookings.
- No booking information needs to persist across reset i.e. once reset is pressed you need not keep track of the bookings anymore.

TA Checkpoints

1. Verify that the student can configure UART for the required baud rate.
2. Verify that the student can demonstrate the UART setup to measure communicate between PC (RealTerm) and Pt-51.
3. Verify that the student has implemented the ATM application with proper working of all 4 states of operation.

Bonus Task (Optional)

As a bonus extension to the Room Booking System, modify your program to introduce an additional **Slot Selection State** between the Date Selection State and the Login Credentials State.

In this enhanced version, each room on each date can be booked in one of the following **three time slots**:

Slot Number	Time Slot
1	8:00 AM – 10:00 AM
2	10:00 AM – 12:00 PM
3	2:00 PM – 5:00 PM

The system must now operate with the following additional state and rules.

1. **Slot Selection State**
   - After a valid room and date are selected, the system should display the following message on the UART terminal:
     > "Select Slot (1–3):"
   - Only the slots that are currently unbooked for the selected **room and date** must be displayed as available options.
   - If the user enters a valid slot number (1–3), that slot should be marked as **booked**, and the system should proceed to the Login Credentials State.
   - If any other input is entered, the following message should be displayed:
     > "**Invalid Slot**"

     The system should return to the Room Selection State.
2. **Availability Rules**
   - A **slot** once booked must not be shown again for the same room and date.
   - If all three slots for a particular date of a room are booked, that date must be removed from the list of available dates for that room.
   - If all dates for a room have all their slots booked, that room must be removed from the list of available rooms.
3. **Summary State (Updated)**
   - Upon successful room selection, date selection, slot selection, and valid login, the UART display should show the following message format:

     ```
     **BOOKING COMPLETE**
     "Room Name" is booked for "Date of Booking"
     Slot: "Time Slot"
     for "Login ID (Lab Name)"
     ```

Implementation Note for Bonus Task:

- Slot availability must be tracked separately for each room and each date.
- Students are expected to extend their data structures appropriately (for example, using a 3D array or equivalent logic).